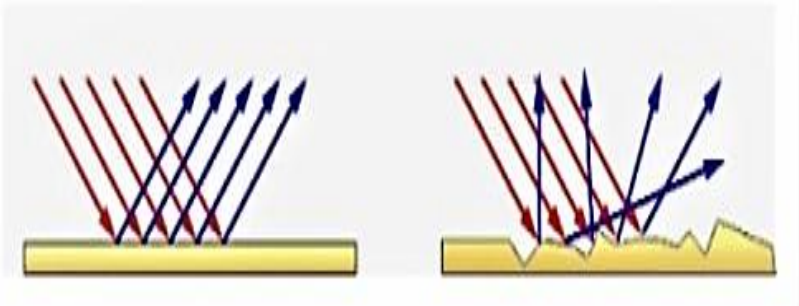
Fizika

9-sinf

Mavzu:

Yorug‘likning qaytish va sinish qonunlari



Yorug‘lik nuri

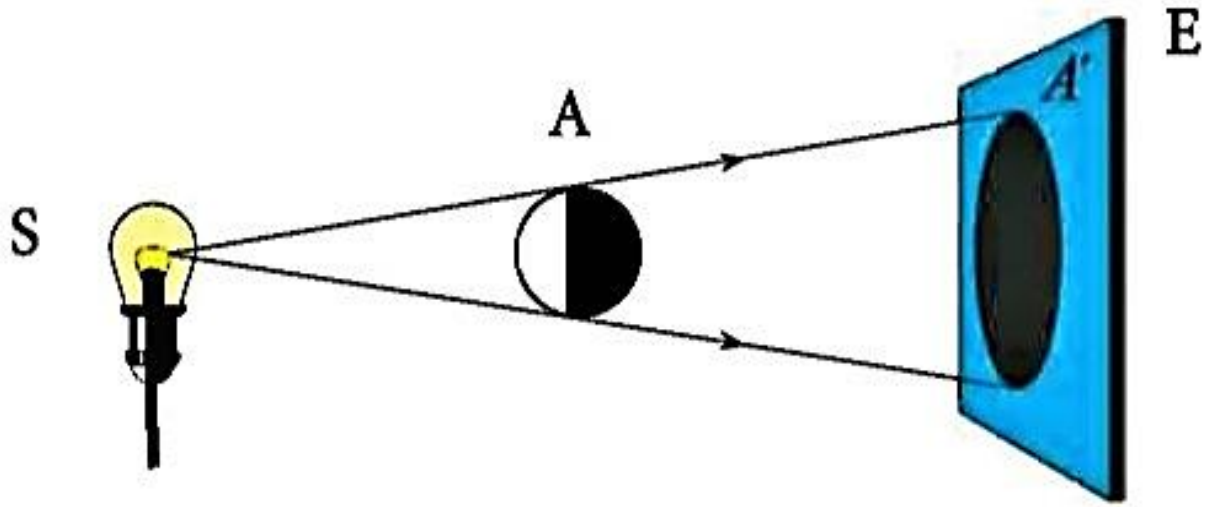
Yo‘nalishlari fazoning ixtiyoriy nuqtasida yorug‘lik energiyasining ko‘chish yo‘nalishi bilan ustma-ust tushgan geometrik chiziq ***yorug‘lik nuri*** deyiladi.

Geometrik optika quyidagi to‘rtta qonunga asoslangan:

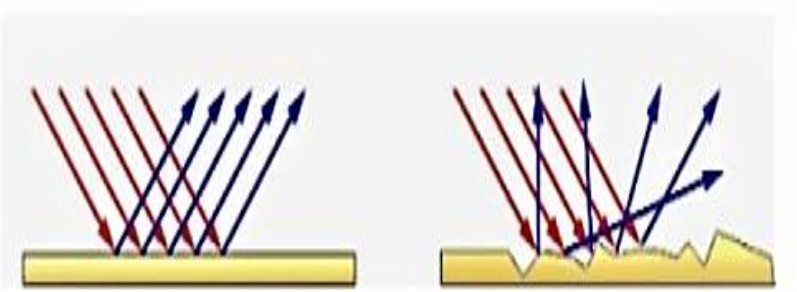
1. Yorug‘likning to‘g‘ri chiziq bo‘ylab tarqalishi;
2. Yorug‘likning mustaqillik qonuni;
3. Yorug‘likning qaytish qonuni;
4. Yorug‘likning sinish qonuni;

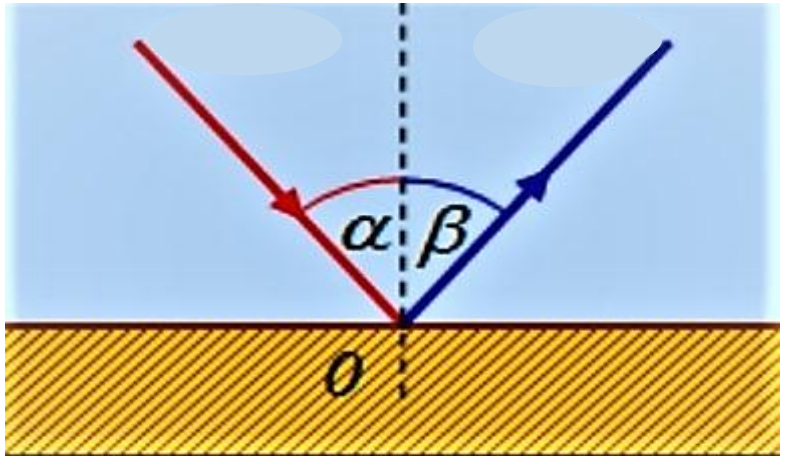
Yorug‘likning to‘g‘ri chiziq bo‘ylab tarqalishi

Soyaning hosil bo‘lishi yorug‘likning to‘g‘ri chiziq bo‘ylab tarqalishini isbotlaydi.



Yorug‘likning qaytishi



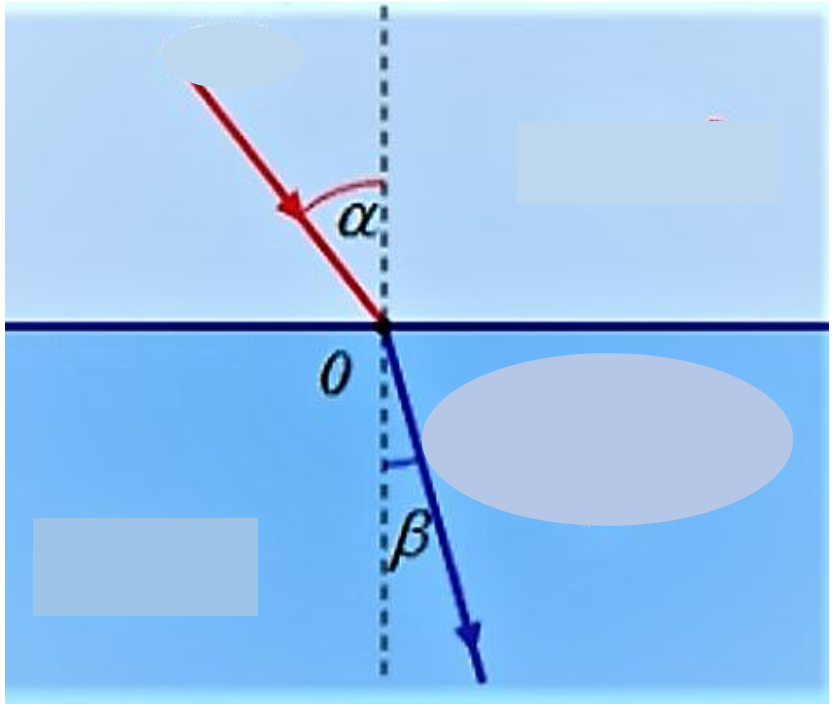


Yorug‘lik g‘adur-budur sirtdan tarqoq (diffuz) qaytadi.

Agar sirt tekis (silliq) bo‘lsa, bunday sirtdan yorug‘lik ko‘zgusimon qaytadi.

1. Tushgan nur, qaytgan nur va tushish nuqtasiga o‘tkazilgan perpendikulyar bir tekislikda yotadi.
2. Qaytish burchagi tushish burchagiga teng: $\alpha = \gamma$

Yorug‘likning sinish qonuni



1. Tushgan nur, singan nur va ikki muhit chegarasiga nurning tushish nuqtasiga o‘tkazilgan perpendikulyar bir tekislikda yotadi.

2. Tushish burchagi sinusining sinish burchagi sinusiga nisbati ikki muhit uchun o‘zgarmas kattalikdir.

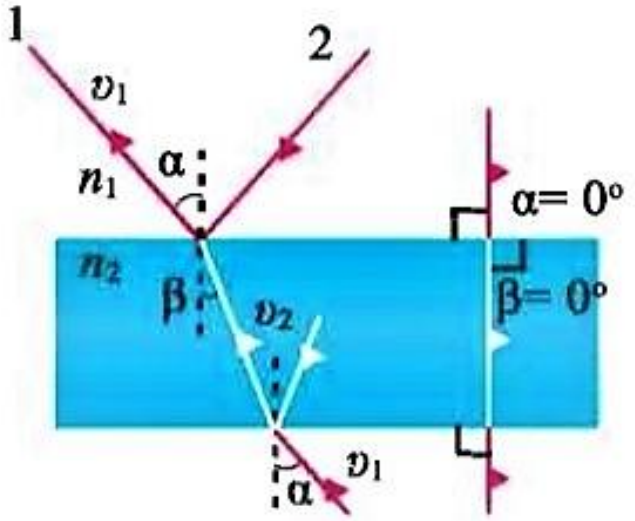
$$n = \frac{\sin \alpha}{\sin \beta}$$

Yorug‘likning sinish qonuni

№	Modda	v, 10^8 m/s	n	№	Modda	v, 10^8 m/s	n
1	Muz	2,29	1,31	4	Kvars	1,95	1,54
2	Suv (20^0 C)	2,25	1,33	5	Yoqut	1,70	1,76
3	Shisha	2,0	1,5	6	Olmos	1,24	2,42

$$n = \frac{c}{\vartheta} \qquad \frac{n_2}{n_1} = \frac{\vartheta_1}{\vartheta_2}$$

$$n_{21} = \frac{n_2}{n_1} = \frac{\sin\alpha}{\sin\beta}$$



$n_2 > n_1$ da $\beta < \alpha$ va $n_2 < n_1$ da $\beta > \alpha$

Masala

Yorug‘lik nurining birinchi muhitdan ikkinchi muhitga o‘tganda tushish burchagi **60º**, sinish burchagi **30º** ga teng.

Ikkinchi muhitning birinchi muhitga nisbatan nur sindirish ko‘rsatkichi qanday?

Berilgan:

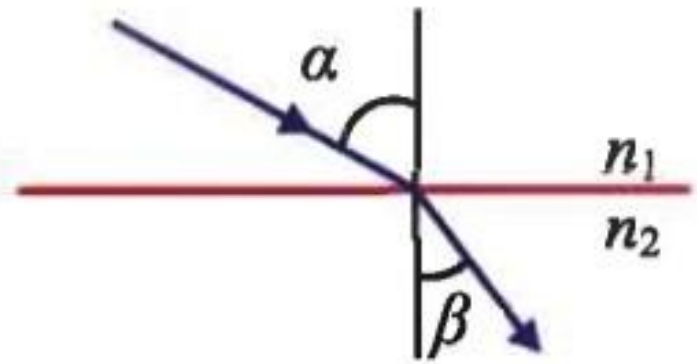
$\alpha = 60°$

$\beta = 30°$

Topish kerak:

$n_{21} = \frac{n_2}{n_1} = ?$

Chizmasi:



Formulasi:

$$n_{21} = \frac{n_2}{n_1} = \frac{\sin\alpha}{\sin\beta}.$$

Hisoblash:

$$n_{21} = \frac{\sin 60^0}{\sin 30^0} = \frac{\frac{\sqrt{3}}{2}}{\frac{1}{2}} = \sqrt{3}.$$

Javob: $n_{21} = \sqrt{3}.$

Mustaqil bajarish uchun topshiriqlar

1. Mavzuni o‘qish va savollarga javob yozish.(133-bet)

2. Stakan ichiga tanga solib, uning ustidan suv quying. Stakandagi suvning sathi orta borishi bilan tanga go‘yoki ko‘tarila borgandek bo‘ladi. Buning sababini tushuntiring.